

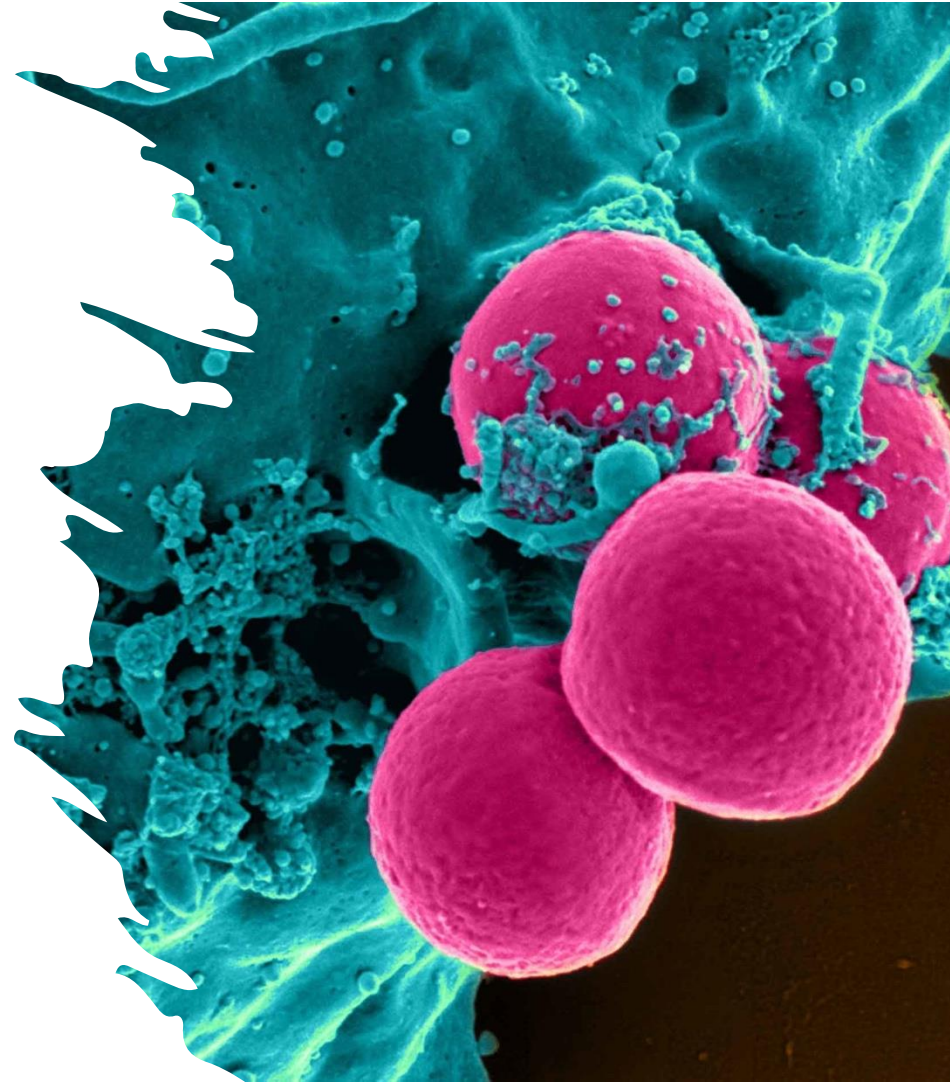
Neoplasia

Dr Nahla Badr

MSc, MD

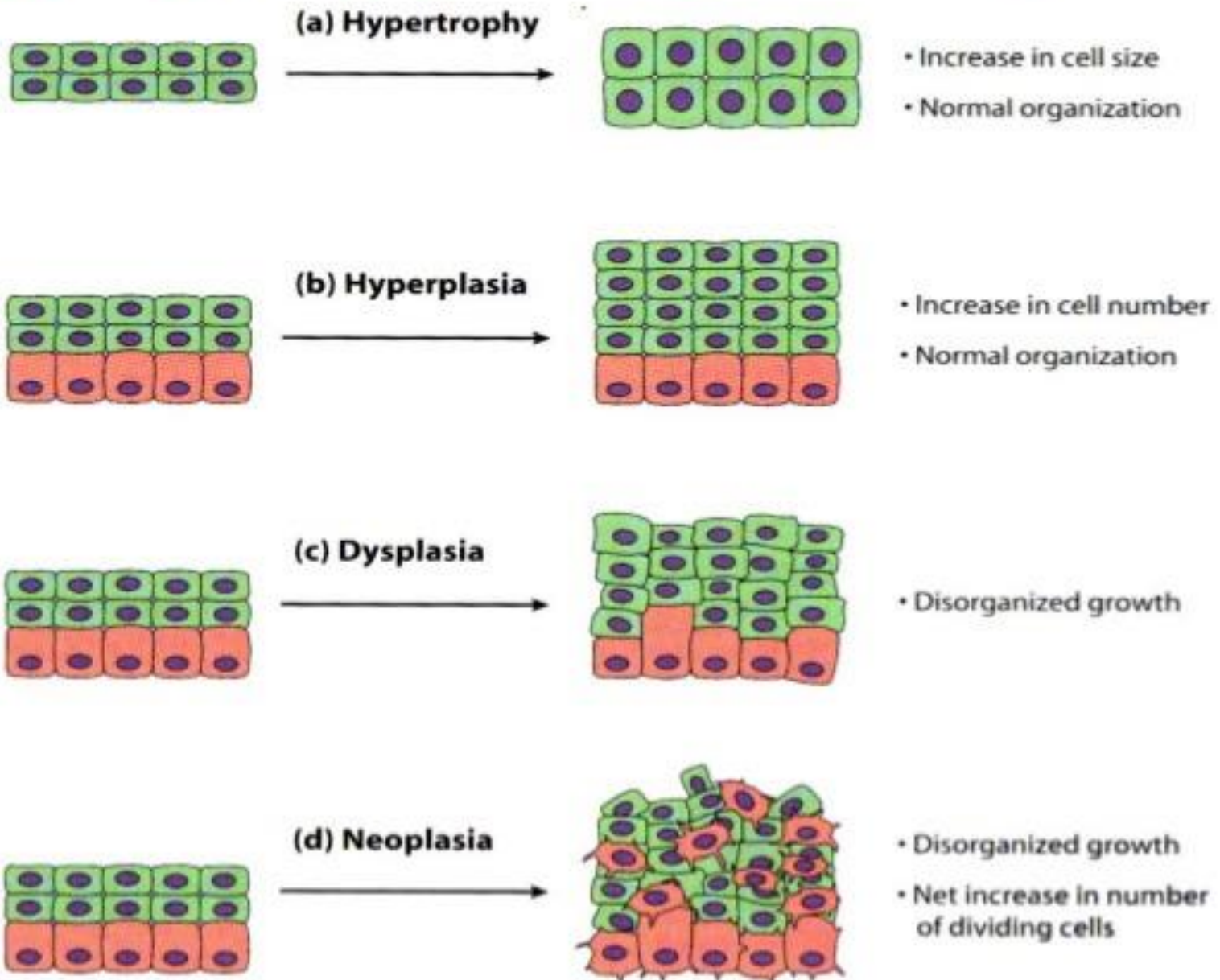
Lecturer of pathology

Menoufia University



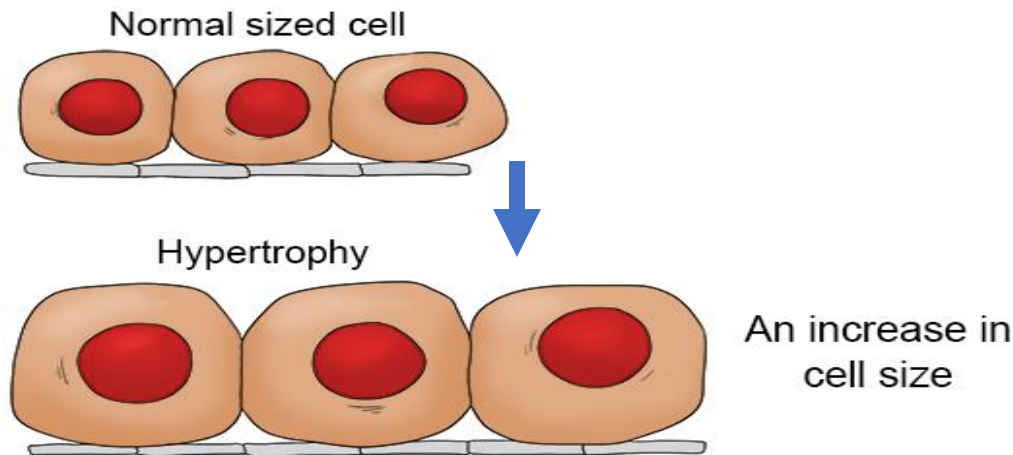
Growth
disturbance

Hypertrophy, hyperplasia, Dysplasia vs Neoplasia



Hypertrophy

- **Definition:** increase in the **size** of cells → increase in the size of the organ (with increased functional capacity)



Hypertrophy

```
graph TD; A[Hypertrophy] --> B[Physiological]; A --> C[Adaptive]; A --> D[Compensatory]; B --> B1[Uterus in pregnancy]; B --> B2[Muscle hypertrophy]; C --> C1[Left ventricular]; C --> C2[Urinary bladder, intestinal wall, stomach hypertrophy]; D --> D1[Kidney hypertrophy];
```

Physiological

Uterus in pregnancy

Muscle hypertrophy

Adaptive

Left ventricular

Urinary bladder,
intestinal wall,
stomach
hypertrophy

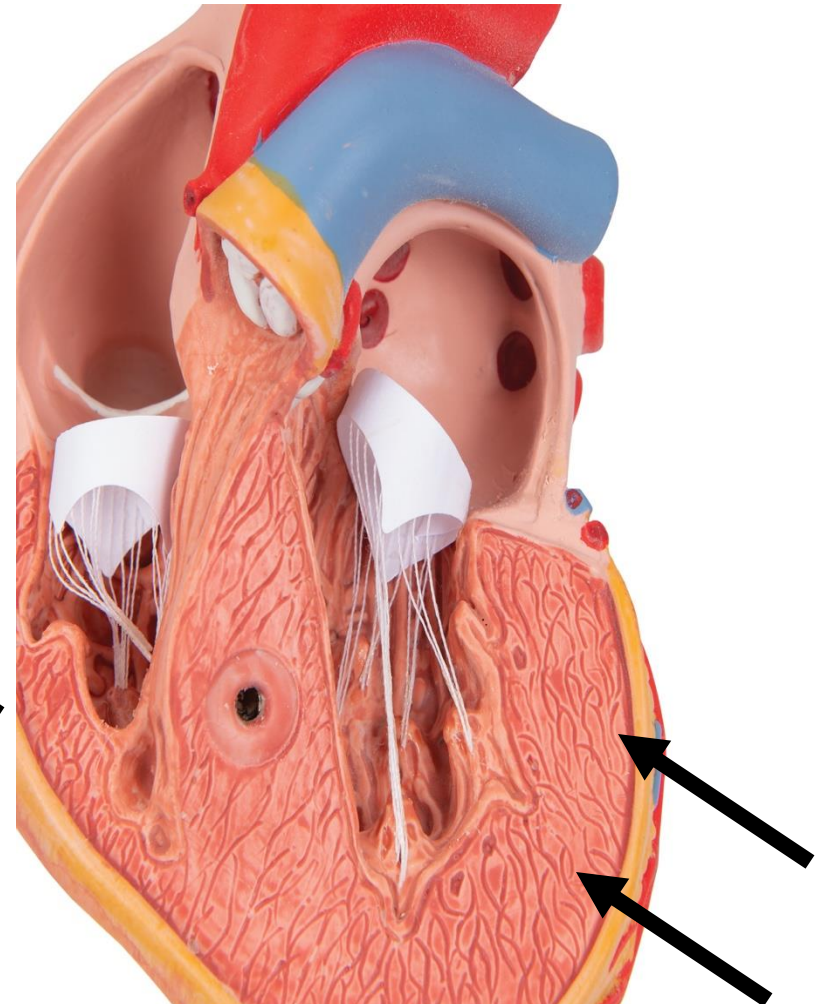
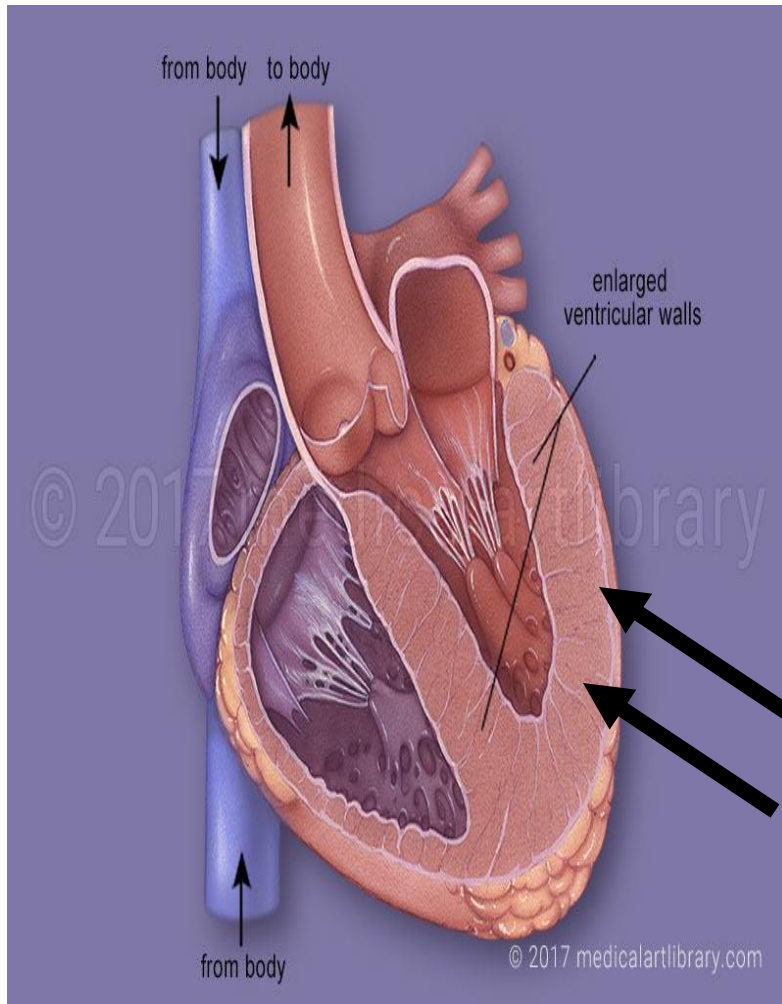
Compensatory

Kidney
hypertrophy

Left ventricular hypertrophy

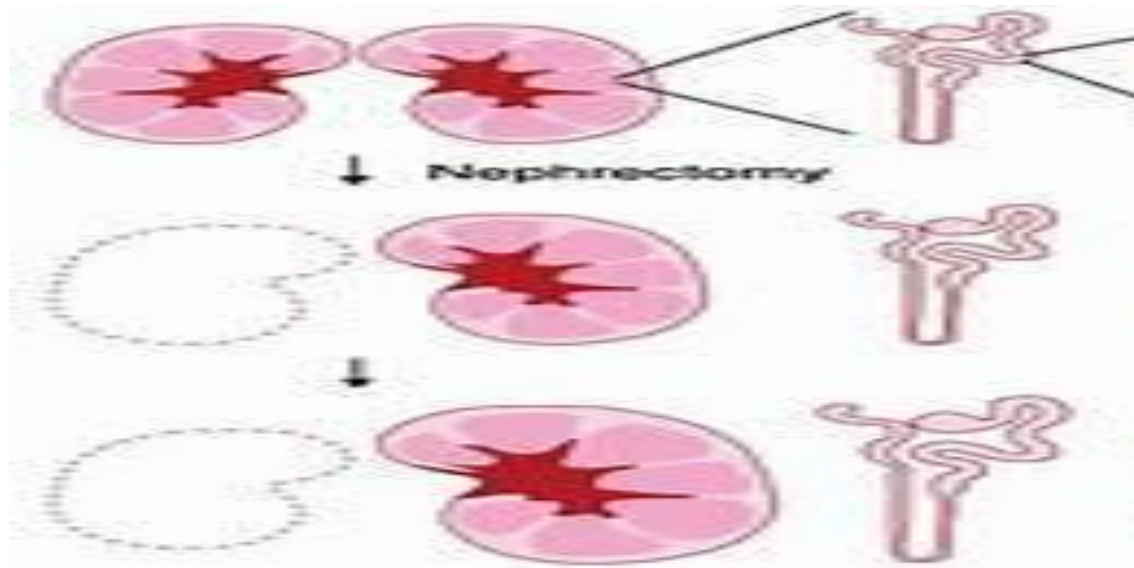
Occurs in hypertension or left sided heart failure.

Left ventricular hypertrophy



Renal hypertrophy

Occurs as a compensatory process after contralateral nephrectomy.

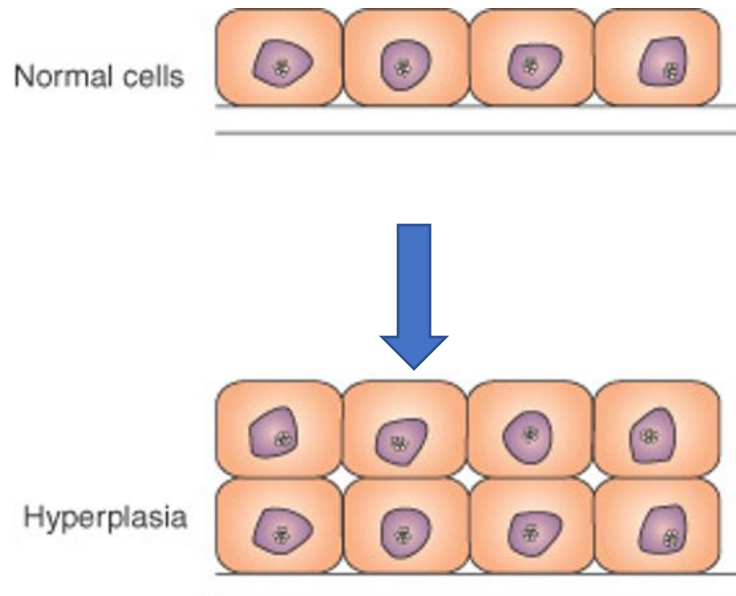
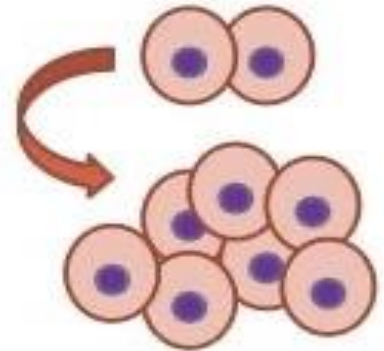


Renal hypertrophy



Hyperplasia

- **Definition:** increase in the **number** of cells
→ increase in the size of the organ.
- It may occur with hypertrophy & often in response to the same stimuli.



Hyperplasia

```
graph TD; A[Hyperplasia] --> B[Physiological]; A --> C[Pathological]
```

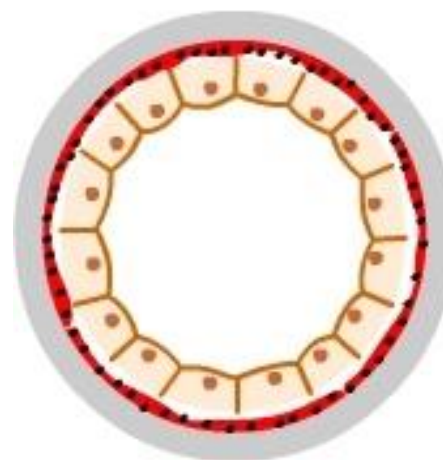
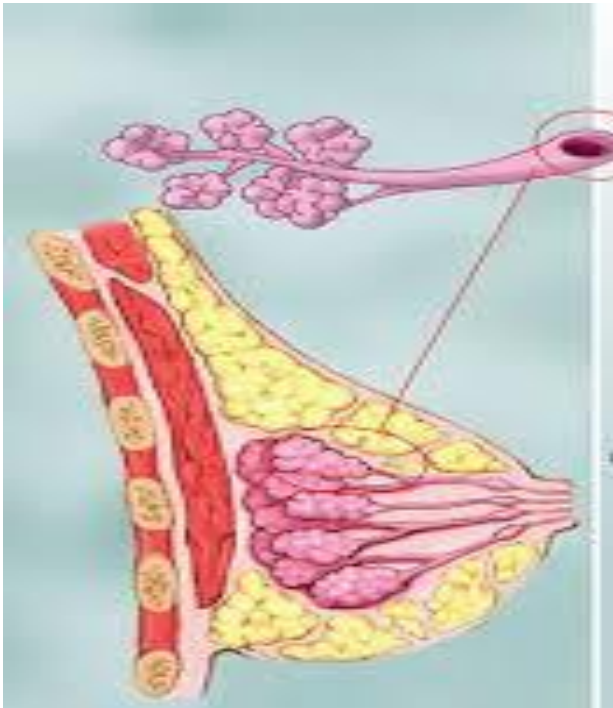
The diagram is a simple hierarchy. At the top is a box labeled 'Hyperplasia'. A vertical line descends from this box and splits into two horizontal lines, each leading to a box below. The left box is labeled 'Physiological' and the right box is labeled 'Pathological'. The boxes are styled with a double-layer effect: an outer orange box and an inner light orange box, both with rounded corners and a thin orange border.

Physiological

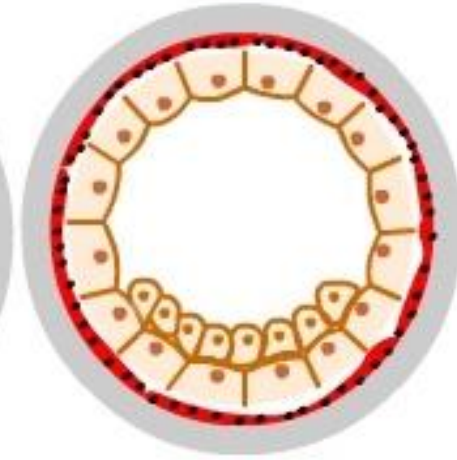
Pathological

Breast hyperplasia

- 1- Physiological: during puberty.
- 2- Pathological: proliferative breast lesions.



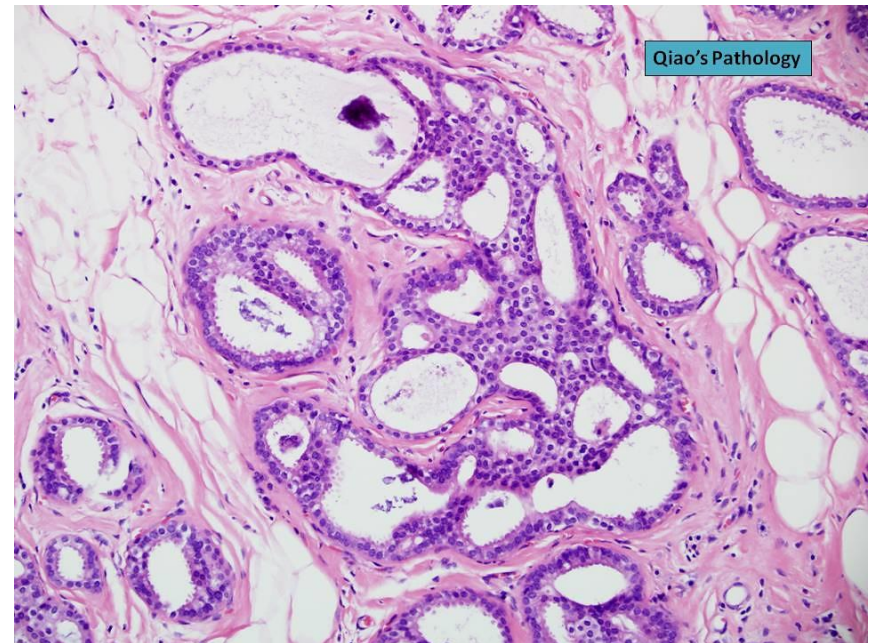
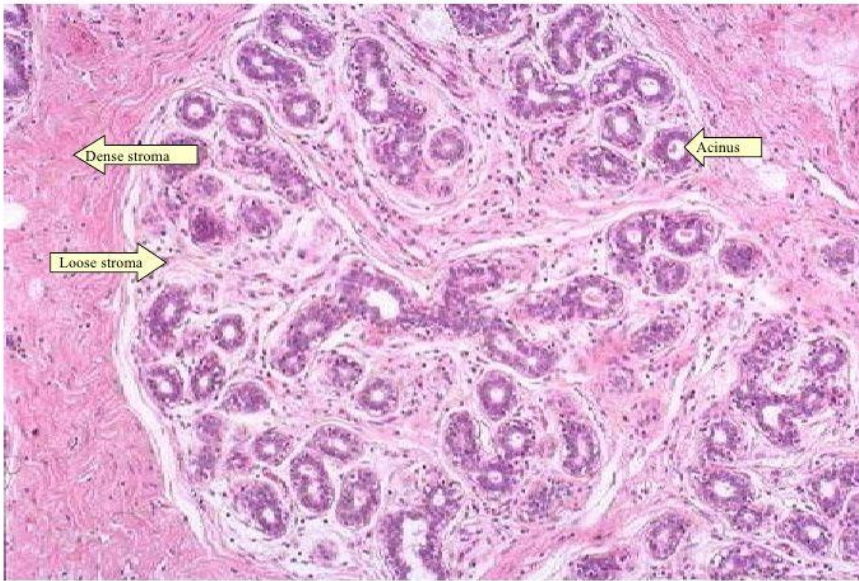
Normal duct



Ductal
hyperplasia

Breast hyperplasia

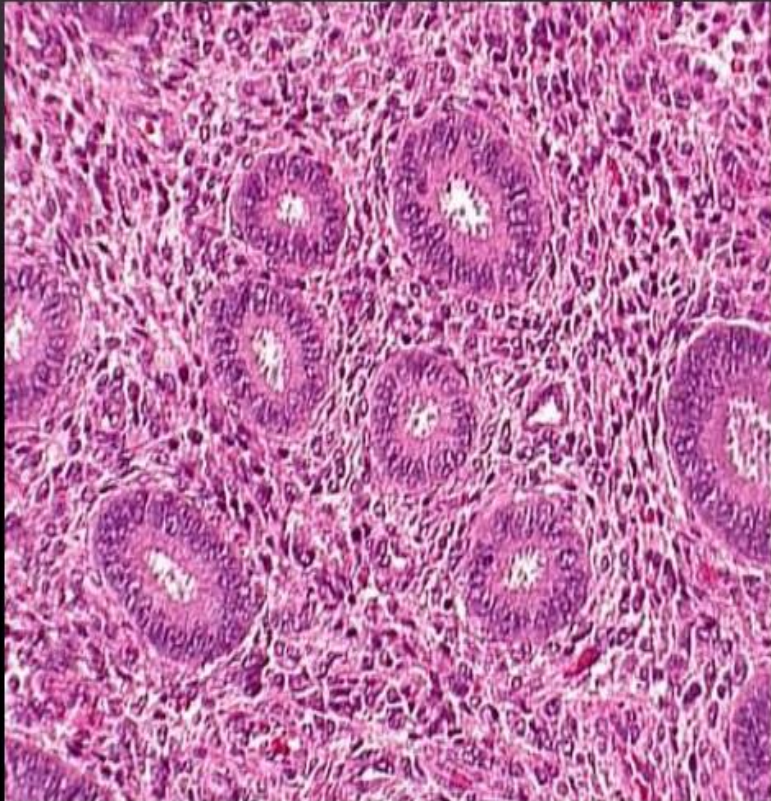
Normal Breast – glands & stroma



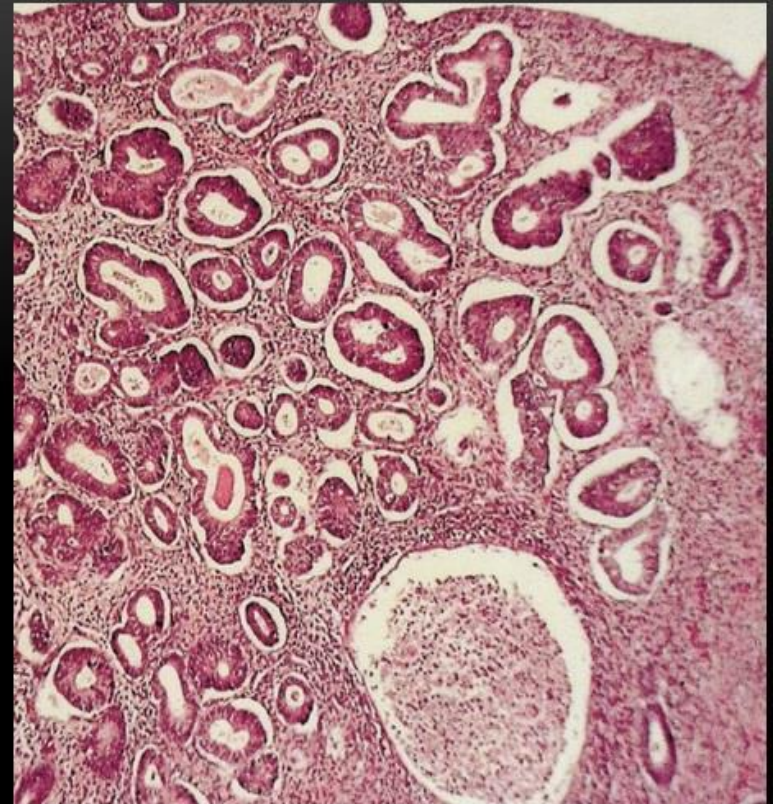
Endometrial hyperplasia

- Pathological: occurs due to increased estrogen hormone level.
- Causes vaginal bleeding.

Endometrial hyperplasia

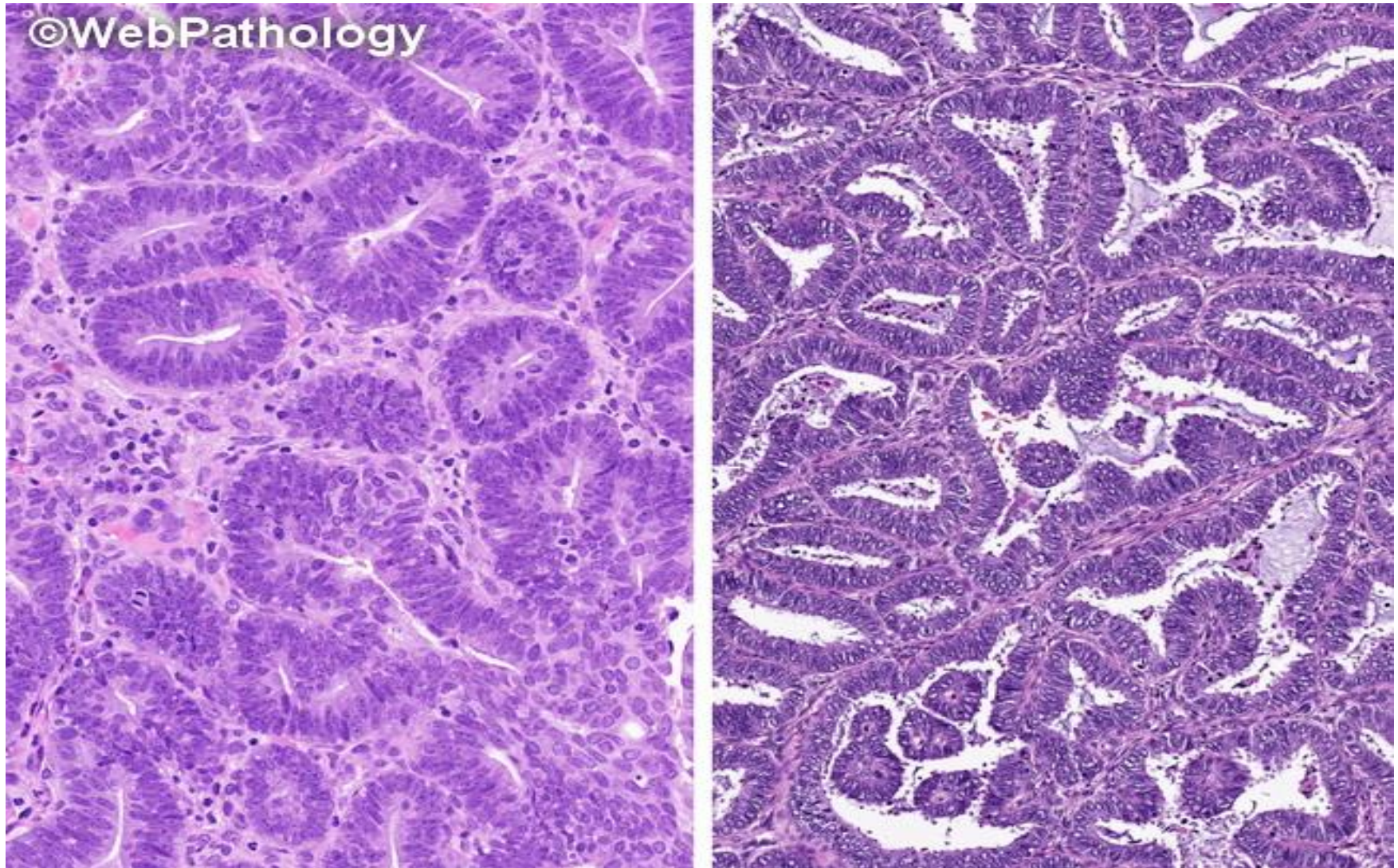


Normal endometrium (Proliferative phase)



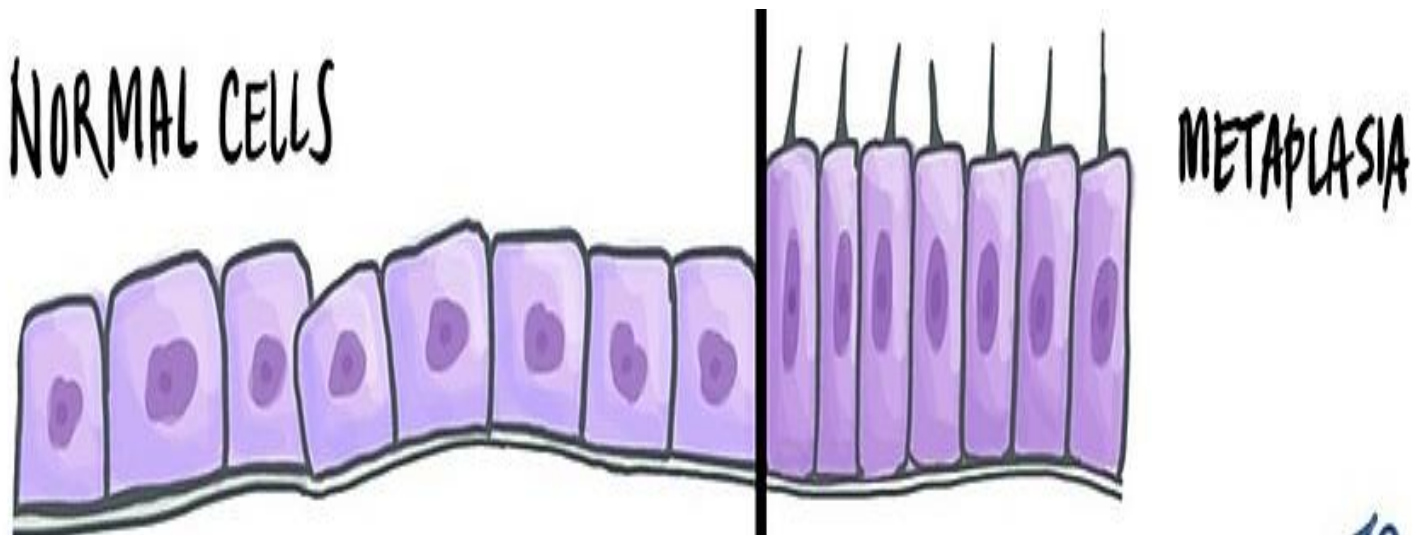
Endometrial Hyperplasia

Endometrial hyperplasia



Metaplasia

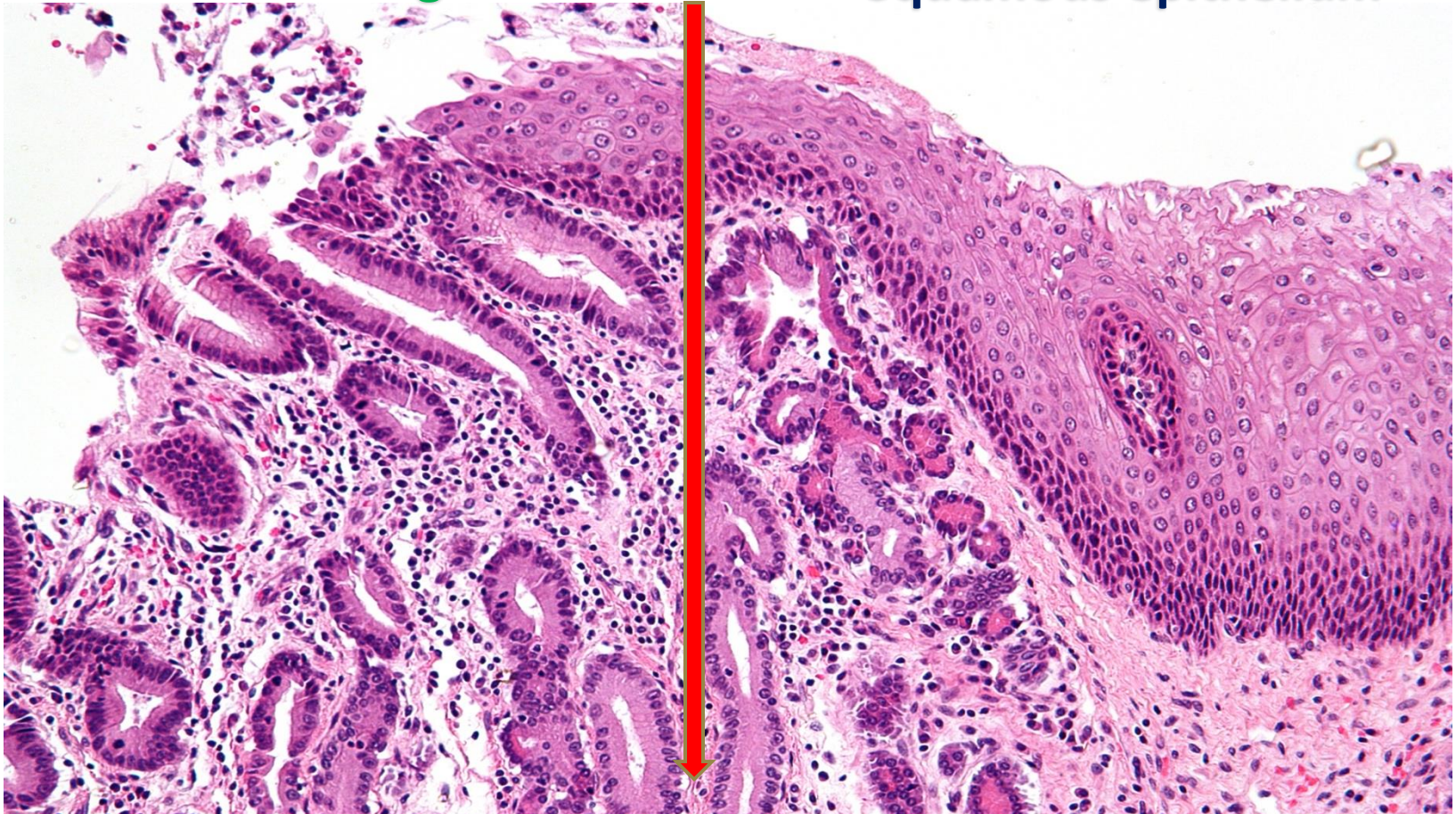
Metaplasia is the transformation of one differentiated cell type to another differentiated cell type.



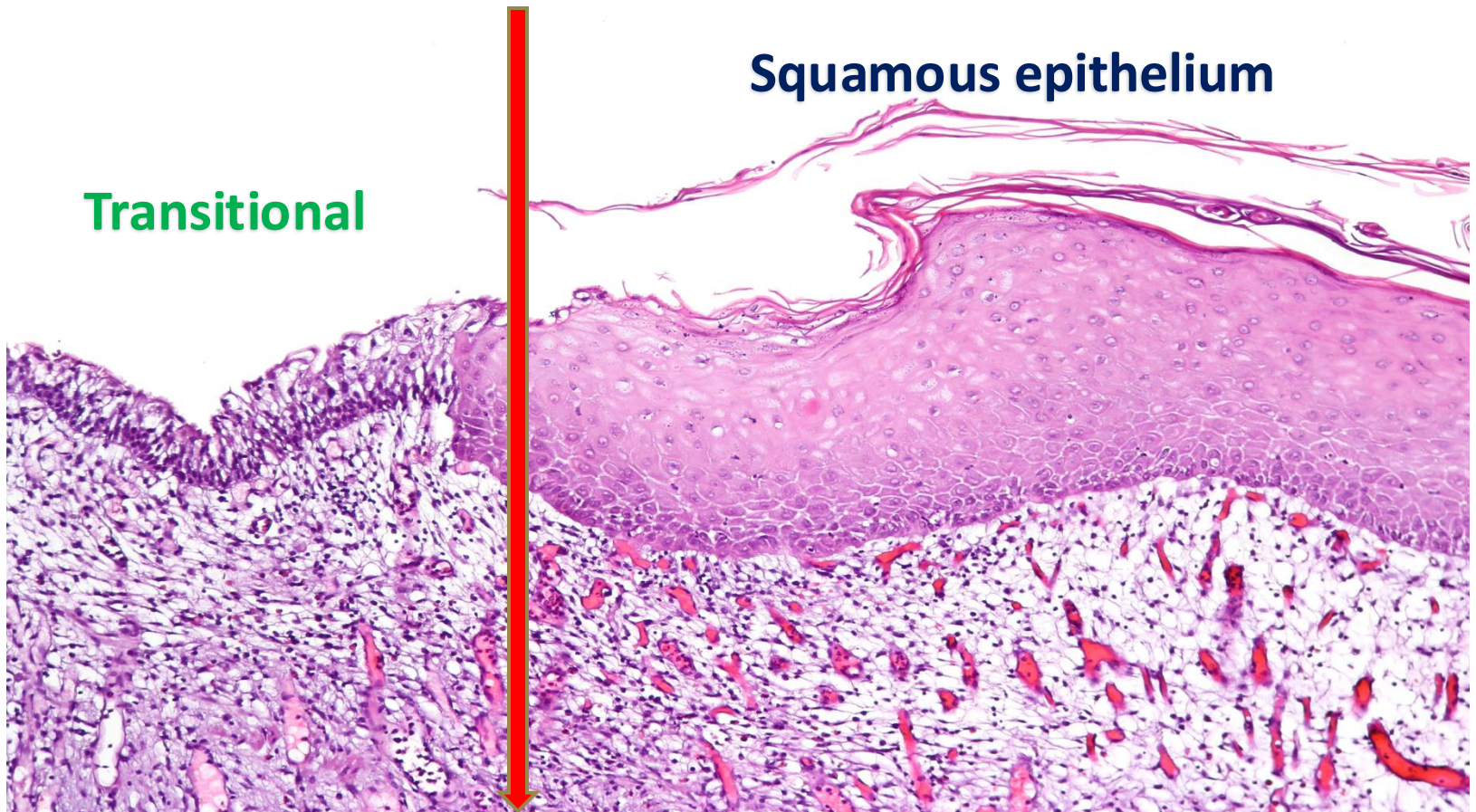
Squamous metaplasia in the wall of the bronchi

Mucous secreting columnar

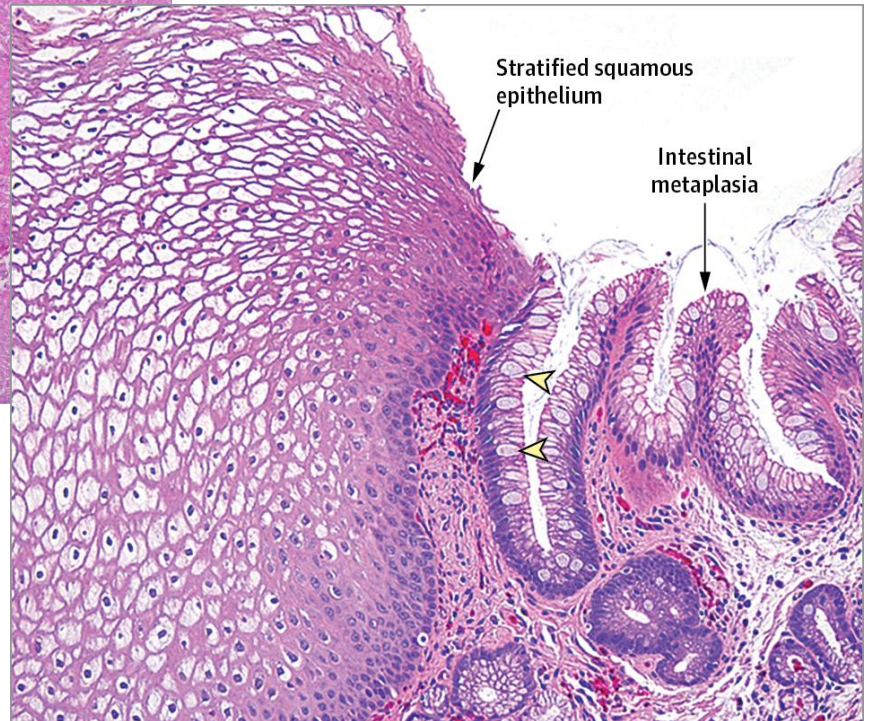
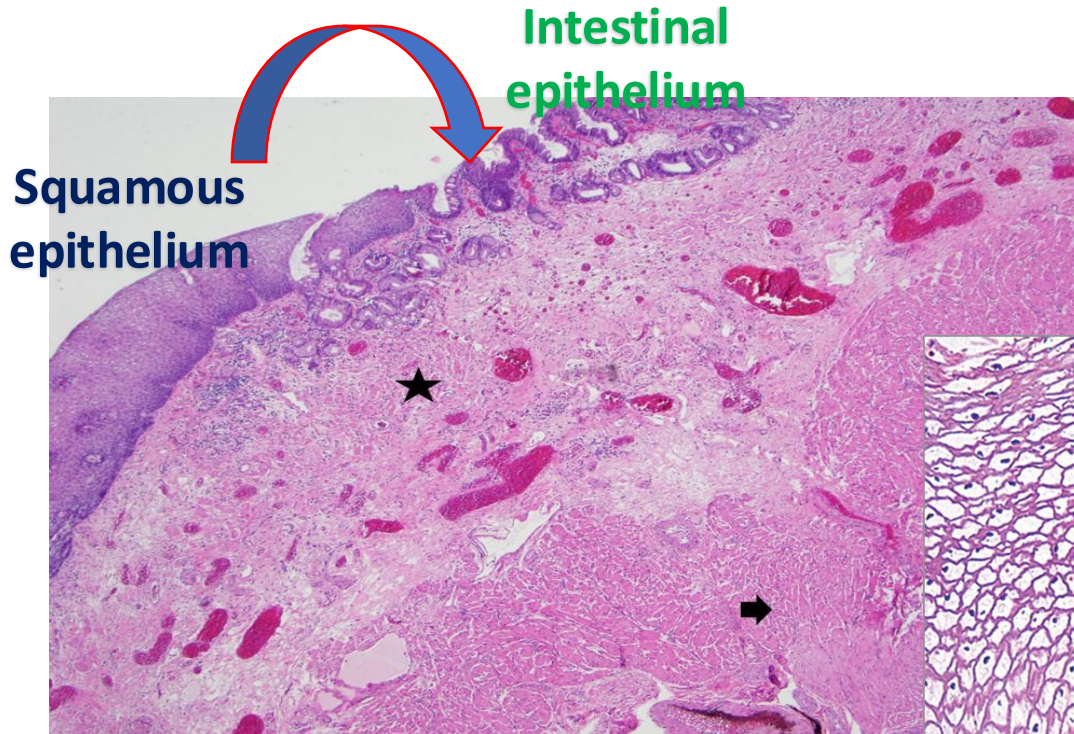
Squamous epithelium



Squamous metaplasia in the urinary bladder



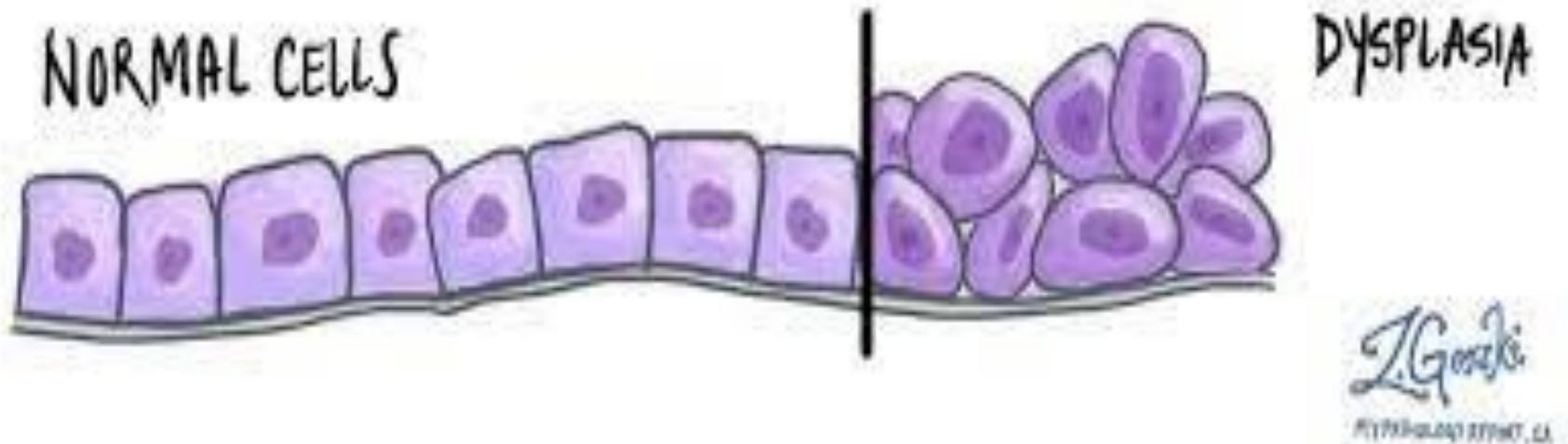
Barrett's esophagus



Dysplasia

Disordered epithelial cell proliferation

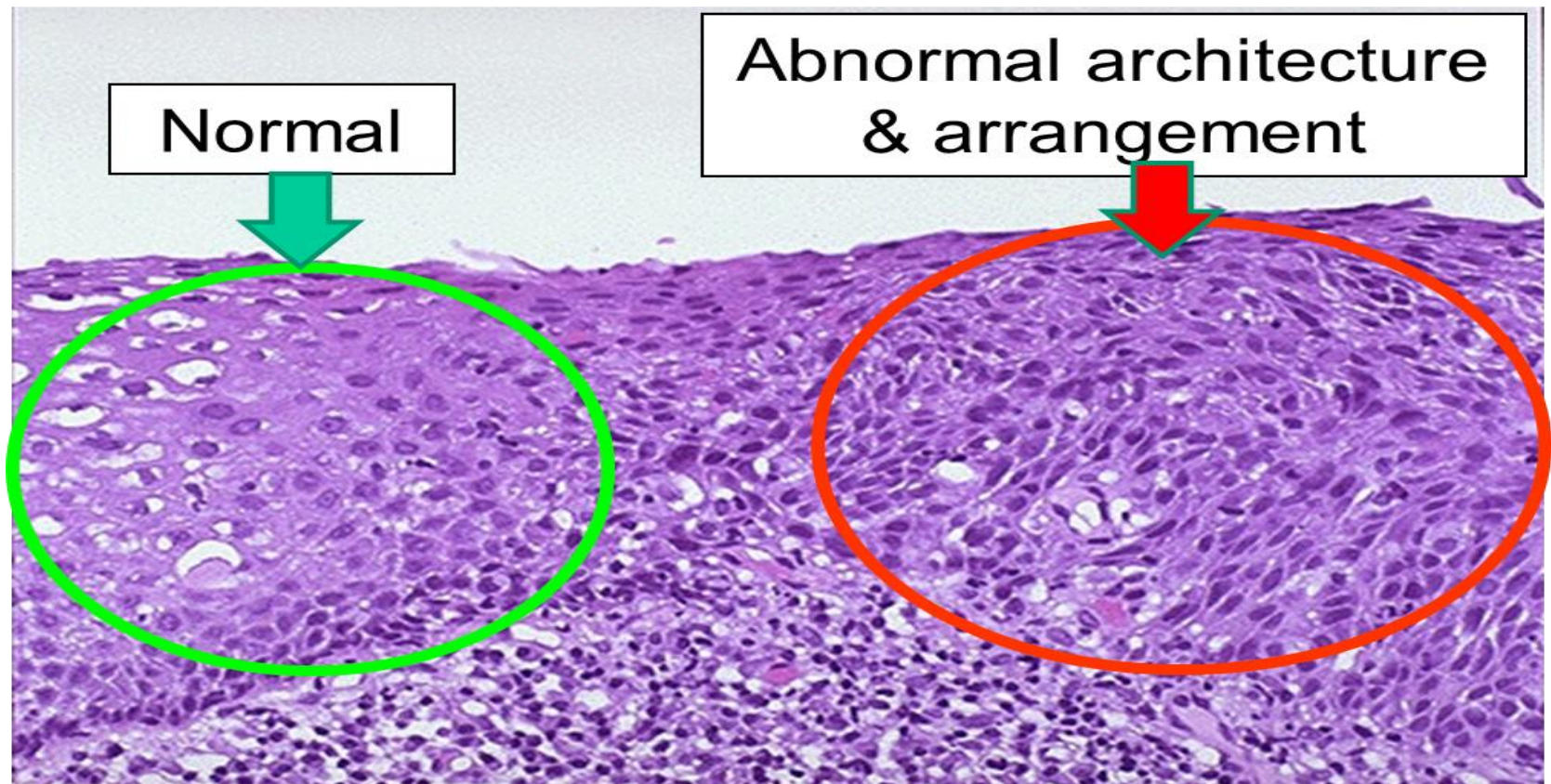
This is **precancerous**



Dysplasia

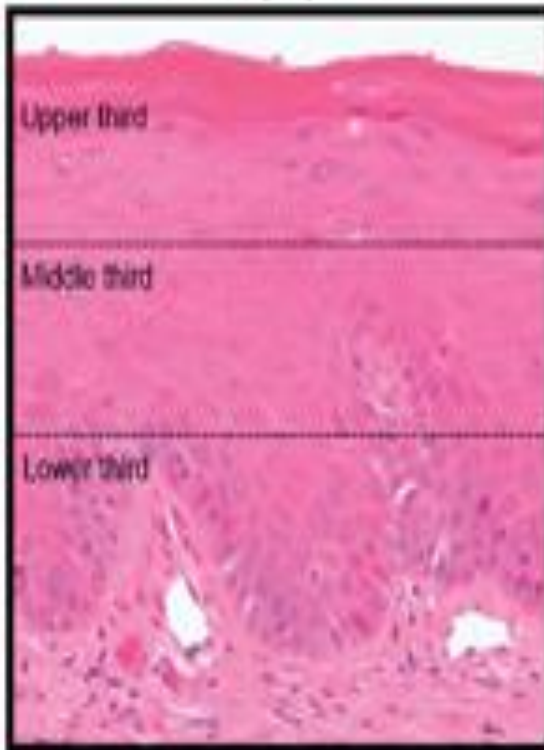
-The Dysplastic cells showed

- Loss of orientation
- Pleomorphism (variation in size & shape)
- Hyperchromasia

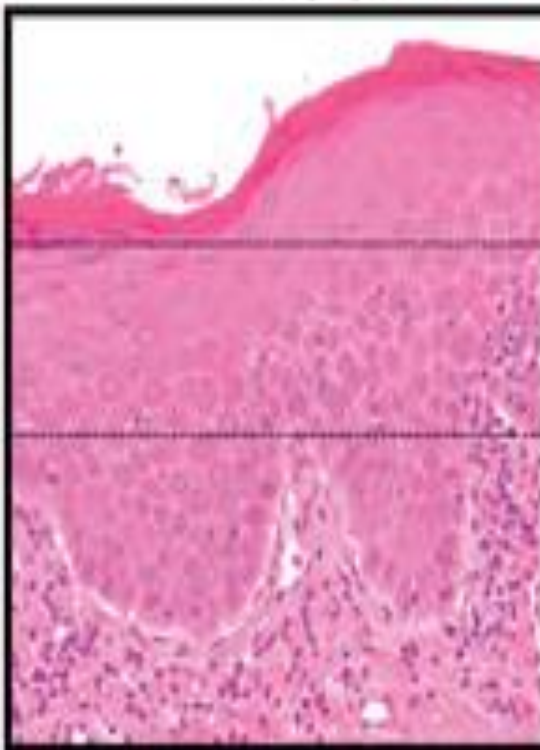


Dysplasia

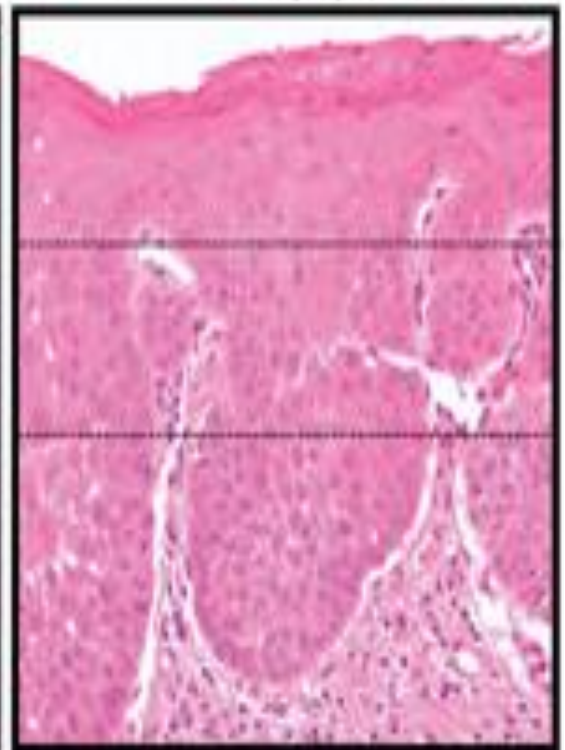
Mild Dysplasia



Moderate Dysplasia

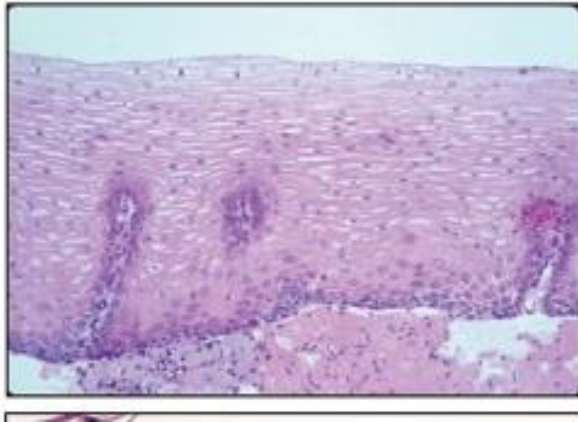


Severe Dysplasia

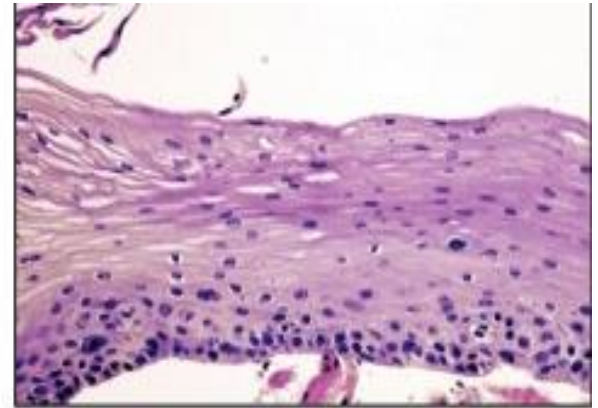


Dysplasia

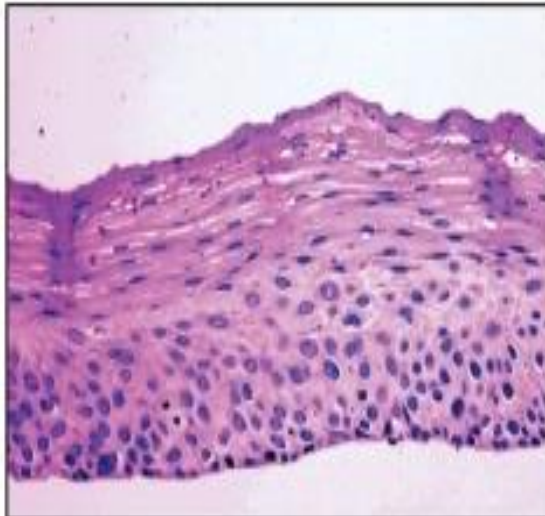
Normal



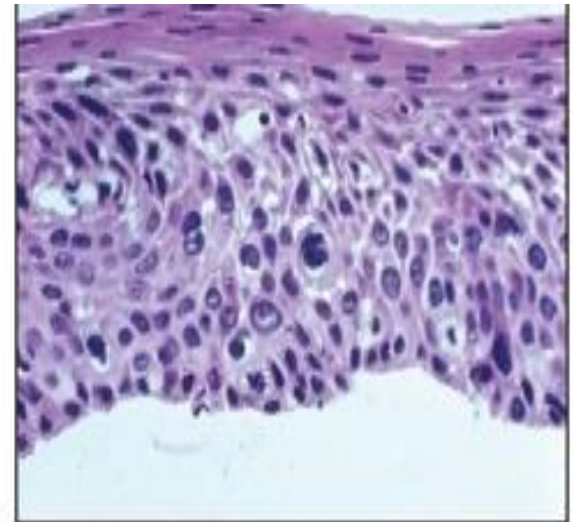
Mild
dysplasia
(LGIN)



Moderate
dysplasia
(MGIN)



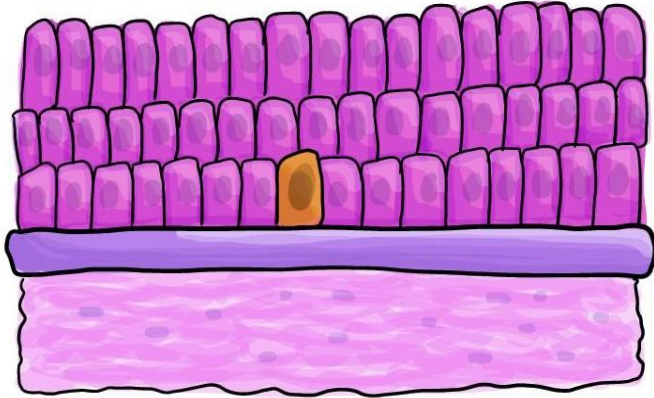
Severe
dysplasia
(HGIN)



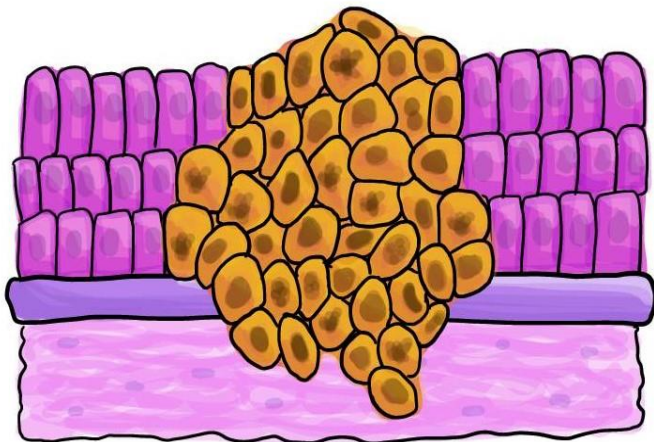
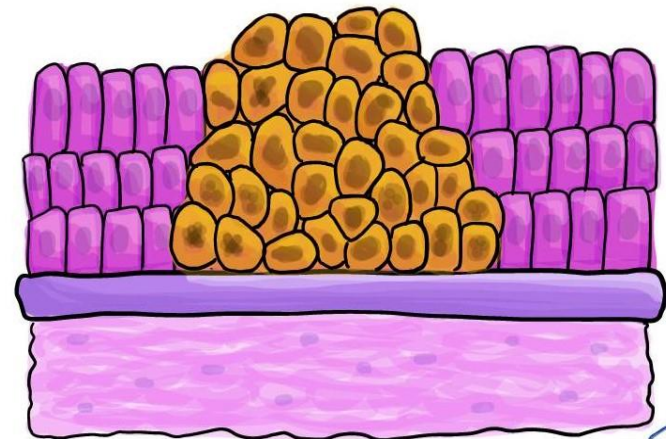
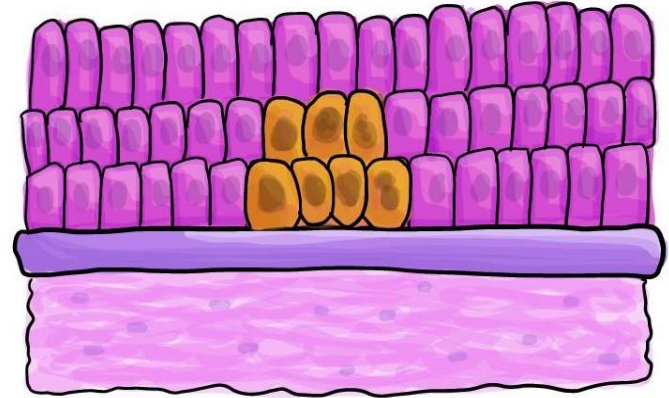
Carcinoma in Situ

- It is sometimes named: Intraepithelial Neoplasia, Intraepithelial Carcinoma or Pre-invasive Carcinoma)
- **Gross picture**: difficult to be detected, some cases may cause a thick indurated patch.
 - **Microscopic picture**:
 1. Cytologic features of atypia (pleomorphism, Hyperchromasia, loss of polarity and abnormal mitosis)
 2. *NO invasion to basement membrane*
 - ***Fate***: Progression into invasive carcinoma occurs after a variable time
 - **Examples**:
 - ***Duct carcinoma in situ in breast (DCIS)***
 - ***Prostatic intraepithelial neoplasia (PIN)***
 - ***Cervical intraepithelial neoplasia (CIN)***

ONE ABNORMAL CELL SURROUNDED
BY NORMAL, HEALTHY CELLS



THE ABNORMAL CELL DIVIDES
TO CREATE MORE ABNORMAL CELLS

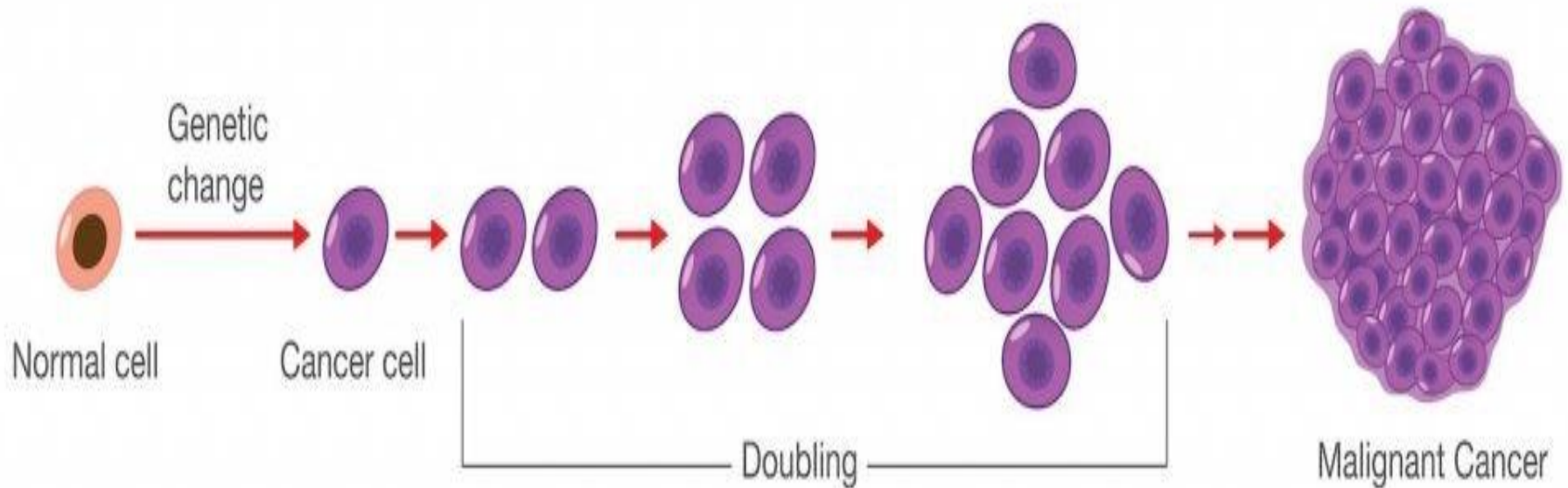


INVASIVE CARCINOMA

CARCINOMA IN SITU

Neoplasia

Process of cancer cell development



Classification of tumors

1- According to behavior:

**Benign ,
malignant and
locally malignant**

2- According to degree of cellular differentiation:

**Well differentiated, moderately differentiated ,poorly
differentiated or undifferentiated**

3- According to tissue of origin:

**Epithelium,
Mesenchymal**

Type of tissue	Origin	Benign tumor	Malignant Tumor
Mesenchymal tissues (Connective tissue)	Fibrous tissue	Fibroma	Fibrosarcoma
	Fatty tissues	Lipoma	Liposarcoma
	Cartilage	Chondroma	Chondrosarcoma
	Bones	Osteoma	Osteosarcoma
	Skeletal muscle	Rhabdomyoma	Rhabdomyosarrcoma
	Smooth muscle	Leiomyoma	Leiomyosarcoma
	Blood veseels	Haemangioma	angiosarcoma
	Lymphatic vessels	lymphangioma	lymphangiosarcoma
Parynchemical tissue	Skin surfaces	Squamous cell papiloma	Squamous cell carcinoma
	Skin	Basal cell papiloma	Basal cell carcinoma
	Gland	Adenoma	Adenocarcinoma

Benign vs Malignant Tumors

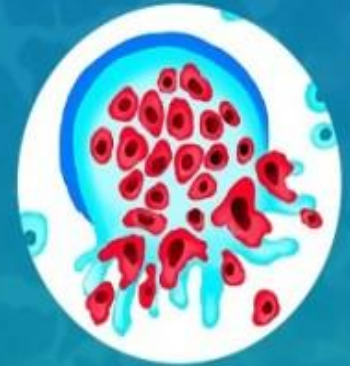
Benign

Non-cancerous, slow-growing,
don't spread (e.g., Lipomas, Fibroids)



Malignant

Cancerous, aggressive,
can spread to other organs



Diagnosis

Imaging (CT, MRI), Endoscopy, Biopsy

Treatment

Benign

Monitoring or removal if needed

Malignant

Surgery, chemotherapy,
radiation, targeted therapies

Regular monitoring is essential for both.

THANK
YOU

